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Medical | IIT-JEE | Foundations

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MM : 720 **Project Phoenix for Re-NEET-2026_CST-05P** **Time : 180 Min.**

Topics Covered:
Complete Syllabus of Class XI & XII

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are three parts in the question paper consisting of Physics, Chemistry and Biology which have 45 questions each in Physics & Chemistry and 90 questions in Biology.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. Which of the following is the dimensional formula for the coefficient of self-induction '*L*'?

(1) $[ML^2T^{-2}A^{-1}]$

(2) $[MT^{-2}A^{-1}]$

(3) $[MLT^{-2}A^{-2}]$

(4) $[ML^2T^{-2}A^{-2}]$

2. For a particle moving along *x*-axis, acceleration is given as $a = 6v^3$, where *a* is in m s^{-2} and *v* is in m s^{-1}. If the velocity of the particle is 2 m s^{-1} at $x = 0$, the velocity at $x = 2$ m is

(1) $\frac{-2}{23}$ m s^{-1}

(2) $-\frac{1}{23}$ m s^{-1}

(3) 23 m s^{-1}

(4) $-\frac{23}{2}$ m s^{-1}

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3. A particle move along the parabolic path, $x = y^2 + 2y + 2$ in such a way that the y-component of velocity vector remain 5 m/s during the motion. The magnitude of the acceleration of the particle is

(1) 50 m/s^2

(2) 100 m/s^2

(3) $10\sqrt{2}$ m/s^2

(4) 0.1 m/s^2

4. A bullet of mass $10\,g$ is fired horizontally with a speed of $400\,m/s$ into a stationary wooden block of mass $1.2\,kg$ resting on a rough horizontal surface. If the coefficient of kinetic friction between the block and the surface is 0.25 then how far will the block-bullet system slide nearly?

(1) 4.2 m

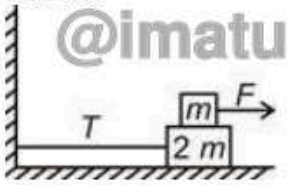
(2) 1.2 m

(3) 3.6 m

(4) 2.2 m

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5. Two blocks of masses m and $2m$ are placed one over the other as shown in figure. The coefficient of friction between m and $2m$ is μ and between $2m$ and ground is $\frac{\mu}{3}$. If a horizontal force F is applied on upper block and tension T is developed in the string, then which of the following is incorrect?



(1) If $F = \frac{\mu mg}{2}, T = 0$

(2) If $F = \frac{\mu}{3}mg, T = 0$

(3) If $F = \mu mg, T = 0$

(4) If $F = 2\mu mg, T = \frac{\mu mg}{3}$

6. A block is sliding down a smooth inclined plane that makes an angle θ with the horizontal. At what value of θ will the horizontal component of the block's acceleration be maximum?

(1) 30°

(2) 45°

(3) 60°

(4) 75°

7. A : Work done by static friction may be negative, positive or zero.
R : The coefficient of static friction in general is larger than coefficient of kinetic friction.

(1) Assertion is true statement but Reason is false

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Both Assertion and Reason are false statements

(4) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

8. A projectile of mass 2 kg is thrown at an angle θ = 60° with the horizontal with speed 100 m/s. At the highest point of projectile it explodes into two equal parts. One part moves vertically upward with speed 100 m/s with respect to the ground. Kinetic energy of other part immediately after the explosion is

(1) 10^4 J

(2) 2×10^4 J

(3) 10^3 J

(4) Zero

9. Two identical balls A and B collides head on elastically. If velocities of A and B were 1 m/s and – 2.5 m/s respectively, then their velocities after the collision are respectively

(1) 1 m/s, – 2.5 m/s

(2) Zero, Zero

(3) – 2.5 m/s, 1 m/s

(4) – 1 m/s, 0.5 m/s

10. Given below are two statements:
Statement A: Moment of inertia of a circular disc about an axis tangential to the rim and lying in the plane of the disc is $\frac{1}{5}MR^2$.
Statement B: Moment of inertia of a circular disc about an axis tangential to the rim and perpendicular to the plane of disc is $\frac{2}{3}MR^2$.
In light of above statements, choose the most appropriate answer among the following options.

(1) Statement A is true but statement B is false

(2) Statement B is true but statement A is false

(3) Both the statements A and B are true

(4) Both the statements A and B are false

11. If a rigid body is in mechanical equilibrium then

(1) There is no change in its linear momentum with time

(2) There is no change in its angular momentum with time

(3) There is no change in its angular momentum but linear momentum can change with time

(4) Both (1) and (2)

12. At what height above the Earth's surface the acceleration due to gravity is 4% of its value on the surface of the Earth? (Radius of the Earth = R)

(1) 2R

(2) 4R

(3) 3R

(4) R

13. **Assertion (A):** A body cannot be shielded from the gravitational influence of nearby matter by putting it inside a hollow sphere.
Reason (R): Gravitational force depends on the medium.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

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14. A rod of uniform cross-sectional area A and length L has a weight W. It is suspended vertically from a fixed support. If Young's modulus for rod is Y, then elongation produced in rod is



(1) $\frac{WL}{YA}$

(2) $\frac{WL}{2YA}$

(3) $\frac{3WL}{4YA}$

(4) $\frac{WL}{4YA}$

15. With the rise in temperature, the viscosity of a liquid and gas changes respectively as

(1) Decreases, increases

(2) Decreases, decreases

(3) Increases, increases

(4) Increases, decreases

16. A tank is filled with water to a height H. A hole is made in one of the wall at a depth D below the water surface. The distance x from the foot of the wall at which the stream coming out of the tank strikes the ground is given by

(1) $x = 2\,[D\,(H–D)]^{1/2}$

(2) $x = 2(g\,D)^{1/2}$

(3) $x = 2\,[D\,(H + D)]^{1/2}$

(4) x = Zero

17. In the spectrum of a black body at two temperature 27°C and 327°C, let A_1 and A_2 be the areas under the two curves respectively. The value of $\frac{A_2}{A_1}$ is

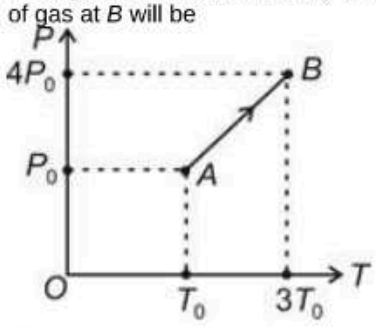
(1) 1 : 16

(2) 4 : 1

(3) 2 : 1

(4) 16 : 1

18. Pressure versus temperature graph for a given sample of an ideal gas is as shown. Density of gas at A is ρ, then density of gas at B will be



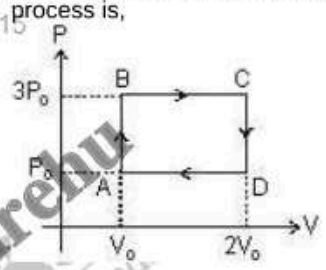
(1) 2ρ

(2) $\frac{1}{2}\rho$

(3) $\frac{1}{4}\rho$

(4) $\frac{4}{3}\rho$

19. An ideal monoatomic gas undergoes a thermodynamic cycle ABCDA as shown in the figure. The efficiency of the cyclic process is,



(1) $\frac{4}{31}$

(2) $\frac{4}{21}$

(3) $\frac{2}{31}$

(4) $\frac{2}{21}$

20. The ratio of molar specific heats $\frac{C_P}{C_V} = \gamma$ can be expressed in terms of degree of freedom (f) as

(1) $\left(1+\frac{f}{3}\right)$

(2) $\left(1+\frac{2}{f}\right)$

(3) $\left(1+\frac{f}{2}\right)$

(4) $\left(1+\frac{1}{f}\right)$

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21. A simple pendulum has time period T_1. The point of suspension is now moved upward according to the relation $y = kt^2$ (k = 2 m/s^2) where y is the vertical displacement. The time period now becomes T_2 then the ratio of $\frac{T_1^2}{T_2^2}$ is (g = 10 m/s^2)

(1) 6 : 5

(2) 5 : 6

(3) 5 : 7

(4) 7 : 5

22. In resonance tube for determination of velocity of sound, two successive positions of resonance are obtained at 15 cm and 48 cm. If the frequency of the fork is 500 Hz, the velocity of sound is

(1) 330 m/s

(2) 300 m/s

(3) 1000 m/s

(4) 360 m/s

23. A simple harmonic progressive wave is represented by the equation $y = 8\sin 2\pi(0.1x - 2t)$, where x and y are in cm and t is in second. At any instant the phase difference between two particles separated by 2.0 cm in the x direction is

(1) 54°

(2) 18°

(3) 72°

(4) 36°

24. The rupture of air medium occurs at $E = 6 \times 10^6$ V/m. The maximum charge that can be given to a uniformly charged sphere of diameter 4 m will be

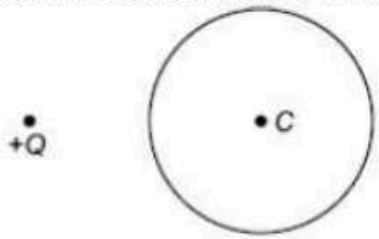
(1) $\frac{4}{3} \times 10^{-3}$ C

(2) $\frac{8}{3} \times 10^{-3}$ C

(3) 4×10^{-3} C

(4) 8×10^{-3} C

25. A point charge $+Q$ is kept near a metallic shell. The electric field at the centre of shell due to charge $+Q$ is



(1) Zero

(2) Rightward

(3) Leftward

(4) Upward

26. An elementary particle of mass m and charge $+e$ is projected with velocity v towards a much more massive particle of charge Ze from a large distance, where $Z > 0$, the closest possible distance between the two particle will be

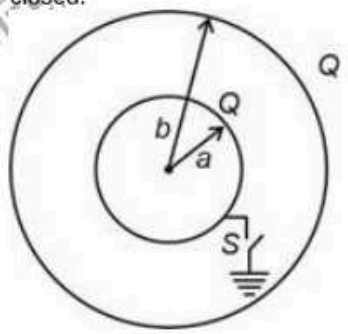
(1) $\frac{Ze^2}{2\pi\varepsilon_0 mv^2}$

(2) $\frac{Ze^2}{8\pi\varepsilon_0 mv^2}$

(3) $\frac{Ze}{8\pi\varepsilon_0 mv^2}$

(4) $\frac{2e}{4\pi\varepsilon_0 mv^2}$

27. Initially, both concentric hollow conducting shells of radii a and b have same charge 'Q' shown in figure and the switch is open. Find the flow of charge through the switch when it is closed.



(1) $Q\left(\frac{a+b}{a}\right)$

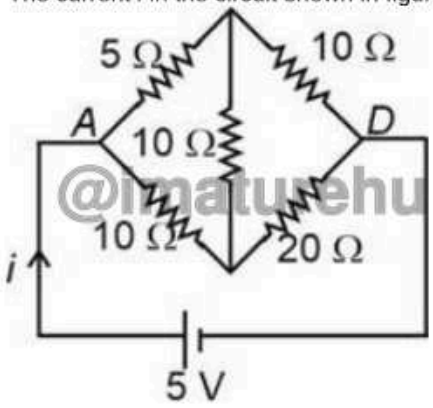
(2) $Q\left(\frac{a+b}{b}\right)$

(3) $\frac{Qa}{b}$

(4) Q

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28. The current *i* in the circuit shown in figure is



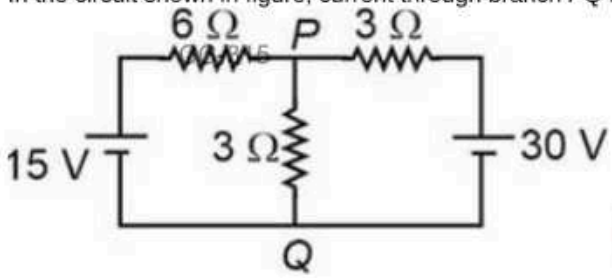
(1) 0.5 A

(2) 0.2 A

(3) 1 A

(4) 2 A

29. In the circuit shown in figure, current through branch *PQ* is



(1) 2 A

(2) 3 A

(3) 4 A

(4) 5 A

30. A galvanometer can be converted into a voltmeter by connecting

(1) High resistance in series with galvanometer

(2) Low resistance in series with galvanometer

(3) High resistance in parallel with galvanometer

(4) Low resistance in parallel with galvanometer

31. A wire carrying current *i*, is kept in *x* – *y* plane along the curve $y = A\sin\left(\frac{2\pi x}{\lambda}\right)$. A uniform magnetic field *B* exists in the *Z*- direction. The magnitude of magnetic force on the portion of the wire between *x* = 0 and $x = \frac{\lambda}{2}$ is

(1) $2Bi\lambda$

(2) $\frac{Bi\lambda}{3}$

(3) $Bi\lambda$

(4) $\frac{Bi\lambda}{2}$

32. Substance in which magnetic moment of a single atom and specimen both are zero, in the absence of magnetising field are known as

(1) Paramagnetic

(2) Ferromagnetic

(3) Diamagnetic

(4) Both 1 and 2

33. A rectangle of side length *L* and breadth *b* lies in the *x*-*y* plane. Magnetic field in the space given by $\vec{B} = B_0\left(2\hat{i} + 3\hat{j} + 4\hat{k}\right)$ T where B_0 is constant. The magnitude of magnetic flux passing through the rectangle is (*L* and *b* are in meter)

(1) 4 B_0Lb Wb

(2) 3 B_0Lb Wb

(3) 2 B_0Lb Wb

(4) Zero

34. If V_{AB} = 10 V and current 1 A is flowing in circuit which increases with the rate of 10^3 A/s, then charge on capacitor is



(1) 2 μC

(2) 7 μC

(3) 14 μC

(4) 10 μC

35. Two coils have a mutual inductance 0.001 H. The current changes in the first coil according to the equation $I = I_o\cos\omega t$, where $I_o = 10A$ and $w = 100\pi\ rad\ s^{-1}$, what is the maximum value of emf in the second coil?

(1) 2π volts

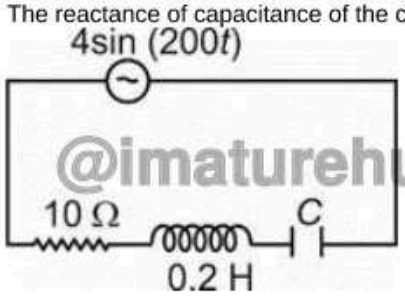
(2) 5π volts

(3) 0 volt

(4) π volt

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36. The power factor of the circuit shown in the figure is $\frac{1}{\sqrt{2}}$. The reactance of capacitance of the circuit is



(1) 30 Ω

(2) 50 Ω

(3) 40 Ω

(4) Either (1) or (2)

37. A plane electromagnetic wave travels in free space along the *x*-direction. The electric field component of the wave at a particular point of space and time is E = 6 Vm^{-1} along *y*-direction. Its corresponding magnetic field component, B would be

(1) 2×10^{-8} T along *y*-direction

(2) 6×10^{-8} T along *z*-direction

(3) 2×10^{-8} T along *z*-direction

(4) 6×10^{-8} T along *x*-direction

38. The critical angle for a denser-rarer interface is 30°. The speed of light in rarer medium is 3×10^8 m /s. The speed of light in denser medium is

(1) 1.1×10^8 m/s

(2) 1.5×10^8 m/s

(3) 2.0×10^8 m/s

(4) 2.5×10^8 m/s

39. If E is the energy of nth orbit of hydrogen atom, the energy of nth orbit of He^+ ion will be

(1) E

(2) $2E$

(3) $3E$

(4) $4E$

40. Two identical glass $\left(\mu = \frac{3}{2}\right)$ equiconvex lenses of focal length f are kept in contact. The space between the lenses is filled with water $\left(\mu = \frac{4}{3}\right)$. The focal length of combination is

(1) f

(2) $\frac{f}{2}$

(3) $\frac{4f}{3}$

(4) $\frac{3f}{4}$

41. When a transparent sheet of thickness t and refractive index μ is introduced in one of the paths in YDSE having distance between the slits as d and distance between slit and screen is D, then the fringe pattern shifts by

(1) $\frac{d}{D}(\mu - 1)t$

(2) $\frac{d}{D(\mu-1)t}$

(3) $\frac{D}{d}(\mu - 1)t$

(4) $\frac{dt}{(\mu-1)D}$

42. The photoelectrons emitted from the surface of metal are such that they

(1) Have same kinetic energy

(2) Have same velocity

(3) Have same de-Broglie wavelength

(4) Have their speeds varying from zero to certain maximum value

43. In the nuclear reaction given below, the unknown Y in product is

${}^1_0n + {}^{235}_{92}U \rightarrow Y + {}^{94}_{38}Sr + 2{}^1_0n$

(1) ${}^{142}_{54}$ Xe

(2) ${}^{140}_{50}$ Xe

(3) ${}^{140}_{52}$ Xe

(4) ${}^{140}_{54}$ Xe

44. When a forward bias is applied to a p-n junction, it

(1) Raises the potential barrier

(2) Reduces the majority carrier current to zero

(3) Lower the potential barrier

(4) Increase the depletion width

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45. Match the following and choose the **correct** option.

Column I	Column II
A. $\overline{A+B}$	(P) $\overline{A}+\overline{B}$
B. $\overline{A\cdot B}$	(Q) $A+B$
C. $\overline{\overline{A}+\overline{B}}$	(R) $\overline{A}\cdot\overline{B}$
D. $\overline{\overline{A}\cdot\overline{B}}$	(S) $A\cdot B$

(1) A → R, B → P, C → S, D → Q

(2) A → R, B → P, C → Q, D → S

(3) A → P, B → R, C → S, D → Q

(4) A → P, B → Q, C → R, D → S

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46. 15 g of urea is dissolved in 90 g water, the mole fraction of urea is approximately

(1) 0.20

(2) 0.05

(3) 0.50

(4) 0.02

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47. Consider the following statements :
(I) First line of Lyman series have larger wavelength than second line of Lyman series
(II) The number of degenerate orbitals in 3rd shell in H-atom is 3
(III) 4*s* orbital is filled prior to 3*d* orbital
The incorrect statement(s) is/are

(1) Only II

(2) I, II

(3) Only III

(4) II, III

48. Correct order of ionic radii is

(1) $O^{2-} > N^{3-} > Mg^{2+} > Na^{+}$

(2) $Mg^{2+} > Na^{+} > O^{2-} > N^{3-}$

(3) $N^{3-} > O^{2-} > Na^{+} > Mg^{2+}$

(4) $Na^{+} > Mg^{2+} > O^{2-} > N^{3-}$

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49. Match the list I with list II.

	List I (IUPAC Name of Element)		List II (IUPAC Official Name)
a.	Unnilunium	(i)	Lawrencium
b.	Unnilquadium	(ii)	Bohrium
c.	Unnilseptium	(iii)	Rutherfordium
d.	Unniltrium	(iv)	Mendelevium

Choose the most appropriate match from the options given below:

(1) a-(iii), b-(iv), c-(i), d-(ii)

(2) a-(iv), b-(iii), c-(i), d-(ii)

(3) a-(iv), b-(iii), c-(ii), d-(i)

(4) a-(iii), b-(iv), c-(ii), d-(i)

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50. Consider the following statements about SF_6 molecule

(a) The central atom is sp^3d^2 hybridized.
(b) The maximum number of atoms lie in a plane are five.
(c) Percentage p character in the hybrid orbitals is 50%.
(d) Maximum number of 90° bond angles are 8.
The correct statements are

(1) (a), (c) and (d) only

(2) (a) and (b) only

(3) (a), (b), (c) and (d)

(4) (a), (b) and (c) only

51. If the observed dipole moment of AB molecule is 2.5 D and A–B bond length is 125 pm then percentage ionic character in the molecule AB will be

(1) 35.2%

(2) 41.6%

(3) 28.6%

(4) 22.6%

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52. The entropy change when 1 mole of ideal gas at 27°C is expanded reversibly from 2 litre to 20 litre is

(1) 2 cal K^{-1} mol^{-1}

(2) 4.6 cal K^{-1} mol^{-1}

(3) 4 J K^{-1} mol^{-1}

(4) 4.6 kJ K^{-1} mol^{-1}

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53. Consider the following statements:
(a) If $\Delta_r H° > 0$ and $\Delta_r S° > 0$, then reaction is spontaneous at high temperature.
(b) At equilibrium, $\Delta G = 0$.
(c) Heat capacity and internal energy are extensive properties.
The correct statements are

(1) (a), (b) and (c)

(2) (a) and (c) only

(3) (b) and (c) only

(4) (a) and (b) only

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54. pH of an aqueous solution of 0.1 M CH_3COONa is (pK_a of CH_3COOH = 4.7)

(1) 7

(2) 7.5

(3) 8.85

(4) 11.35

55. Match reaction in List-I with their correct relation of K_p and K_c in List-II

List-I	List-II
(a) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$	(i) $K_p = K_c(RT)^{-2}$
(b) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	(ii) $K_p = K_c(RT)$
(c) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$	(iii) $K_p = K_c$
(d) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$	(iv) $K_p = K_c(RT)^{-1}$

Choose the **correct** match

(1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(2) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(3) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

(4) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

56. Among the following species, the one that can undergo disproportionation reaction is

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(1) ClO_2^-

(2) MnO_4^-

(3) $Cr_2O_7^{2-}$

(4) SO_4^{2-}

57. Equivalent weight of Cl_2 according to given redox reaction will be

$$aCl_2 + bNaOH \rightarrow cNaCl + dNaClO_3 + eH_2O$$

(1) 71 g

(2) 16.2 g

(3) 85.2 g

(4) 42.6 g

58. Identify the **correct** statements from the following.
(a) ZSM-5, a type of zeolite, is used to convert alcohols into gasoline.
(b) CO is highly poisonous in nature.
(c) CO_2 is used as fire extinguisher.
(d) The basic structural unit of silicates is SiO_4^{4-}.

(1) (a) and (b) only

(2) (a) and (c) only

(3) (b) and (c) only

(4) (a), (b), (c) and (d)

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59. In Dumas method for estimation of nitrogen, 0.1 g of an organic compound gave 30 mL of nitrogen collected at 300 K temperature and 715 mm Hg pressure. The percentage composition of nitrogen in the compound is
(Aqueous tension at 300 K = 15 mm Hg)

(1) 19.7%

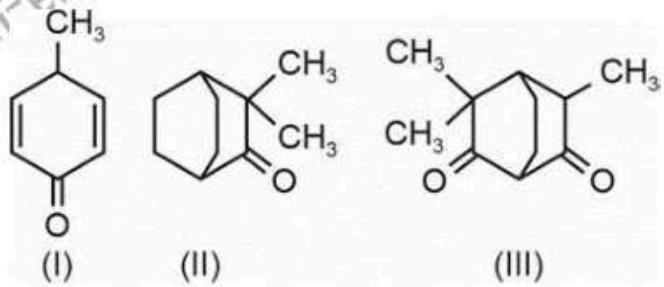
(2) 49.1%

(3) 28.2%

(4) 31.4%

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60.



Which of the given compounds can exhibit tautomerism?

(1) I, II and III

(2) Only I and II

(3) Only I and III

(4) Only II and III

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61. Match List-I with List-II.

List-I (Compound)	List-II (Number of σ-bonds)
(a) Anthracene	(i) 23
(b) Biphenyl	(ii) 18
(c) Gammaxane	(iii) 19
(d) Naphthalene	(iv) 26

Choose the correct match

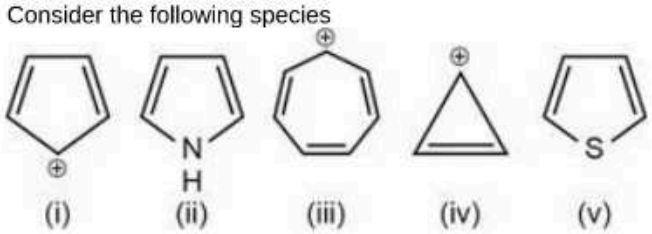
(1) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

(2) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)

(3) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(4) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

62. Consider the following species



(i) (ii) (iii) (iv) (v)

The total number of aromatic species are

(1) 2

(2) 4

(3) 5

(4) 3

63. Consider the following statements.

Statement I: Solubility of gases in liquids increases with increase in temperature.

Statement II: For a solution of volatile liquids, the partial vapour pressure of each volatile component of the solution is directly proportional to its mole fraction in solution.

In the light of above statements, choose the most appropriate answer from the options given below

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is incorrect but statement II is correct

(4) Statement I is correct but statement II is incorrect

64. Vapour pressure of two volatile liquids A and B are 300 mm Hg and 200 mm Hg respectively. The molar mass of B is twice that of A. If equal masses of A and B are mixed, then the mole fraction of B in vapour phase is equal to

(1) 0.75

(2) 0.25

(3) 0.40

(4) 0.60

65. On electrolysis of an aqueous solution of $CuCl_2$ using platinum (Pt) electrode, the product obtained at cathode will be

(1) Hydrogen gas

(2) Oxygen gas

(3) Chlorine gas

(4) Copper metal

66. Match the list I with list II.

List I	List II (Reaction at anode)
a. Mercury cell	(i) $2H_2(g) + 4OH^-(aq) \rightarrow 4H_2O(l) + 4e^-$
b. Dry cell	(ii) $Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4(s) + 2e^-$
c. H_2–O_2 fuel cell	(iii) $Zn(s) \rightarrow Zn^{2+} + 2e^-$
d. Lead storage battery (on discharging)	(iv) $Zn(Hg) + 2OH^- \rightarrow ZnO(s) + H_2O + 2e^-$

Choose the correct match from the options given below:

(1) a(iv), b(iii), c(ii), d(i)

(2) a(iv), b(iii), c(i), d(ii)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(iii), b(iv), c(i), d(ii)

67. In the reaction

$$BrO_3(aq) + 5\,Br^-(aq) + 6H^+(aq) \rightarrow Br_2(l) + 3H_2O(l)$$

If rate of disappearance of Br^- ion is 4.5×10^{-3} molL^{-1} s^{-1}, then rate of appearance of Br_2 is

(1) 9.0×10^{-4} mol L^{-1} s^{-1}

(2) 1.35×10^{-2} mol L^{-1} s^{-1}

(3) 2.7×10^{-3} mol L^{-1} s^{-1}

(4) 1.5×10^{-3} mol L^{-1} s^{-1}

68. A first order of reaction takes 70 minutes for 80% completion. The time taken for 90% completion of reaction will be

(1) 114 minutes

(2) 140 minutes

(3) 100 minutes

(4) 200 minutes

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69. Match the List I with List II

List I (Compounds)		List II (Method of preparation)
a. Metaphosphoric acid	(i)	$P_2O_3 + H_2O$
b. Phosphonic acid	(ii)	$PCl_3 + H_3PO_3$
c. Hypophosphoric acid	(iii)	Red P_4 + alkali
d. Pyrophosphorous acid	(iv)	Phosphorus acid + Br_2, heat in a sealed tube

The correct match is

(1) a(ii), b(iii), c(i), d(iv)

(2) a(iv), b(i), c(iii), d(ii)

(3) a(i), b(ii), c(iii), d(iv)

(4) a(iv), b(i), c(ii), d(iii)

70. Number of lone pairs present on xenon in XeF_6, XeF_4, XeF_2 respectively

(1) 1, 2, 3

(2) 2, 1, 3

(3) 3, 2, 1

(4) 1, 3, 2

71. Match List-I with List-II

	List I (Ion)		List II (Spin only magnetic moment value in BM)
a.	Ni^{2+}	(i)	1.73
b.	Cr^{2+}	(ii)	2.83
c.	Ti^{3+}	(iii)	3.87
d.	Co^{2+}	(iv)	4.90

Choose the correct answer from the options given below

(1) a(ii), b(iv), c(iii), d(i)

(2) a(ii), b(iii), c(i), d(iv)

(3) a(iv), b(iii), c(i), d(ii)

(4) a(ii), b(iv), c(i), d(iii)

72. Correct order of ionic radii of the given species is

(1) $Nd^{3+} > Dy^{3+} > Eu^{3+} > Yb^{3+}$

(2) $Eu^{3+} > Nd^{3+} > Dy^{3+} > Yb^{3+}$

(3) $Yb^{3+} > Dy^{3+} > Eu^{3+} > Nd^{3+}$

(4) $Nd^{3+} > Eu^{3+} > Dy^{3+} > Yb^{3+}$

73. Consider the following statements about $[PtCl_2(en)_2]^{2+}$

(a) cis-$[PtCl_2(en)_2]^{2+}$ is optically active

(b) It has two geometrical isomers

(c) It is diamagnetic in nature

The correct statement(s) is/are

(1) (b) only

(2) (a), (b) and (c)

(3) (a) only

(4) (b) and (c) only

74. Which among the following complexes will show Jahn Teller distortion?

(a) $[FeF_6]^{3-}$

(b) $[CoF_6]^{3-}$

(c) $[Co(C_2O_4)_3]^{3-}$

(d) $[Fe(CN)_6]^{3-}$

Choose the correct option.

(1) (a) and (c) only

(2) (a), (b) and (c) only

(3) (b) and (d) only

(4) (a), (b), (c) and (d)

75. Arrange the following compounds in their increasing order of reactivity towards S_N1 reaction

a. $CH_3CH_2CH_2CH_2 - Br$

b. $CH_3CH_2-\underset{\underset{Br}{|}}{CH}-CH_3$

c. $(CH_3)_3CBr$

(1) b < a < c

(2) a < c < b

(3) a < b < c

(4) c < b < a

76. Consider the following reaction sequence

$$CH_3-\underset{\underset{Br}{|}}{CH}-CH_3 \xrightarrow[\Delta]{\text{Alcoholic KOH}} A \text{ (Major)} \xrightarrow[\text{(ii) } H_2O_2/OH^-/H_2O]{\text{(i) } B_2H_6} B \text{ (Major)} \xrightarrow{CrO_3-H_2SO_4} C \text{ (Major)}$$

Major product C is a/an

(1) Carboxylic acid

(2) Aldehyde

(3) Ketone

(4) Alkene

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77. Consider the following statements:
(a) Phenol on reaction with dilute nitric acid at 298 K yields a mixture of ortho and para nitrophenols.
(b) Ortho and para nitrophenols can be separated by steam distillation.
(c) o-Nitrophenol is steam volatile due to intramolecular hydrogen bonding.
Correct statements are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a), (b) and (c)

(4) (a) and (c) only

78. Phenetole on reaction with HBr majorly gives

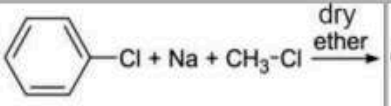
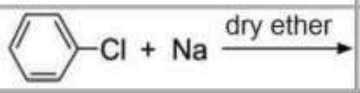
(1) Phenol and ethyl bromide

(2) Benzyl bromide and methyl bromide

(3) Phenol and methyl bromide

(4) Bromobenzene and ethyl bromide

79. Match List-I with List-II

	List-I (Reaction)		List-II (Name of reaction)
(a)		(i)	Fittig reaction
(b)		(ii)	Swarts reaction
(c)	$CH_3 - Cl + NaI \xrightarrow{\text{dry acetone}}$	(iii)	Finkelstein reaction
(d)	$CH_3 - Br + AgF \rightarrow$	(iv)	Wurtz-Fittig reaction

The correct match is

(1) a(i), b(iv), c(ii), d(iii)

(2) a(iv), b(i), c(iii), d(ii)

(3) a(i), b(iii), c(ii), d(iv)

(4) a(i), b(iv), c(iii), d(ii)

80. Which of the following compounds give cannizzaro reaction?
(a) 2-Methylpropanal
(b) Benzaldehyde
(c) Acetophenone
(d) 2,2-Dimethylpropanal
The correct option is

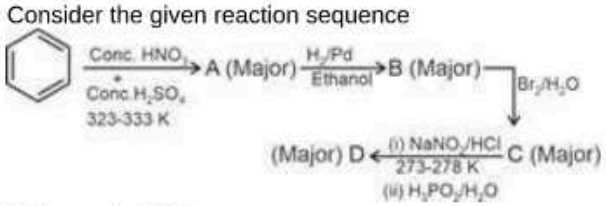
(1) (a), (c) and (d) only

(2) (a), (b) and (d) only

(3) (a) and (b) only

(4) (b) and (c) only

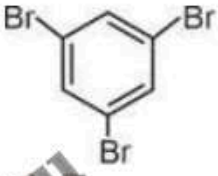
81. Consider the given reaction sequence

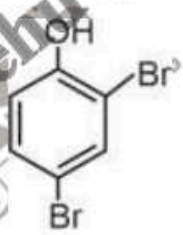


Major product D is

(1) 

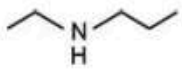
(2) 

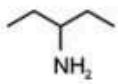
(3) 

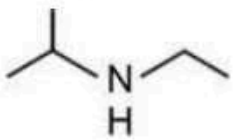
(4) 

82. A compound 'X' having molecular formula $C_5H_{13}N$ on reaction with Hinsberg's reagent gives compound 'Y' which dissolves in aqueous alkali. The structure of the compound 'X' could be

(1) 

(2) 

(3) 

(4) 

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83. Match list I (vitamins) with list II (deficiency diseases)

List I (Vitamins)	List II (Deficiency Diseases)
a. Thiamine	(i) Convulsions
b. Riboflavin	(ii) Beri-beri
c. Pyridoxine	(iii) Scurvy
d. Ascorbic acid	(iv) Cheilosis

Select the correct option.

(1) a(iii), b(i), c(ii), d(iv)

(2) a(iii), b(iv), c(i), d(ii)

(3) a(ii), b(iii), c(i), d(iv)

(4) a(ii), b(iv), c(i), d(iii)

84. In a test tube containing a salt, a few drops of dilute H_2SO_4 was added, which gave colourless vapours having the smell of vinegar. The vapours turned the blue litmus paper red. Identify the **correct** anion from the following :

(1) Acetate, CH_3COO^-

(2) Carbonate, CO_3^{2-}

(3) Sulphate, SO_4^{2-}

(4) Sulphide, S^{2-}

85. Consider the following statements:

(a) Griess-Ilosvay test is the confirmatory test of the nitrite ion

(b) Brown ring test is the confirmatory test of nitrate ion

(c) In qualitative analysis, group reagent for Ba^{2+} and Ni^{2+} cations are $(NH_4)_2CO_3$ in presence of NH_4OH and H_2S gas in presence of dilute HCl respectively

(d) On passing ammonia gas through Nessler's reagent a brown precipitate is formed

(e) Lead sulphate is soluble in ammonium acetate solution due to the formation of ammonium tetraacetoplumbate(II)

The correct statements are

(1) (b), (d) and (e) only

(2) (a), (b), (d) and (e) only

(3) (a) and (c) only

(4) (a), (b), (c), (d) and (e)

86. Which of the following statements are true

(a) Number of stereoisomers of $Ni(PPh_3)_4$ are 2

(b) Number of triangular faces (having one N-atom and two Cl atom) in fac form of $[Co(NH_3)_3Cl_3]$ is 3

(c) Total number of stereoisomers of $[Co(en)_2(ox)]Cl$ is more than that of $[Co(en)_2Cl_2]Br$

(d) $[Co(CO)_4]^-$ is paramagnetic while $[Fe(CO)_4]^{2-}$ is diamagnetic in nature.

Choose the correct option.

(1) a, b and c only

(2) b only

(3) a, b and d only

(4) a and c only

87. 5 ml of a gas containing only carbon and hydrogen were mixed with an excess of oxygen (30 mL) and the mixture exploded by means an electric spark. After the explosion, the volume of mixed gases remaining was 25mL. On adding a concentrated solution of KOH, the volume further diminished to 15 ml. Considering complete combustion of hydrocarbon. The molecular formula of hydrocarbon is (All volumes have been reduced to NTP).

(1) C_3H_8

(2) C_4H_8

(3) C_2H_4

(4) C_5H_8

88. An electron in hydrogen atom in its ground state absorbs two times as much energy as the minimum energy required for it to escape from the atom. The de-Broglie wavelength of emitted electron will be (ionization energy of hydrogen atom = 13.6 eV, $h = 6 \times 10^{-34}$ Js and $M_e = 9 \times 10^{-31}$ kg)

(1) 2.17×10^{-9} m

(2) 3.04×10^{-11} m

(3) 3.04×10^{-10} m

(4) 2.17×10^{-10} m

89. An aqueous solution of a metal bromide MBr_2 (0.046 M) is saturated with H_2S. What is the minimum pH at which MS will precipitate?

(K_{sp} for MS = 6.0×10^{-21} ; concentration of saturated H_2S = 0.1 M, $K_1 = 10^{-7}$ and $K_2 = 1.3 \times 10^{-13}$ for H_2S)

(1) 1

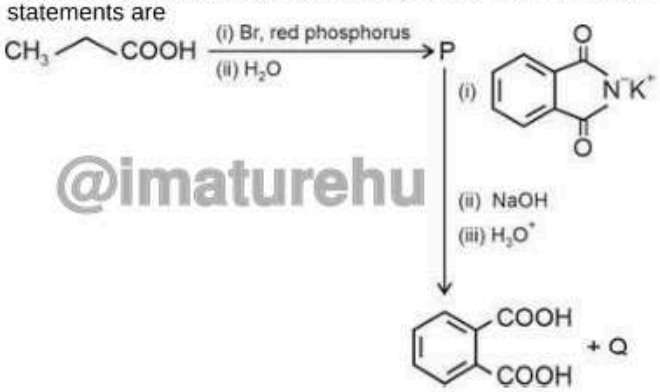
(2) 1.5

(3) 2

(4) 3

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90. Considering the reaction sequence given below, the correct statements are



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(a) Treating 'P' with conc. NH_4OH solution followed by acidification gives 'Q'

(b) pKa of 'P' is more than CH_3CH_2COOH

(c) Treating 'Q' with a solution of $NaNO_2$ in aq. HCl liberates N_2

(d) 'P' can be reduced to a primary alcohol using BH_3·THF.

Choose the **correct** option

(1) (a) and (c) only

(2) (a), (c) and (d) only

(3) (a) and (b) only

(4) (b), (c) and (d) only

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BIOLOGY

91. Mark the **incorrect** statement w.r.t. binomial nomenclature.

(1) Three words collectively form binomial epithet

(2) The name of discoverer is printed in italics

(3) Specific epithet starts with a small letter

(4) Biological names are generally taken from Latin language

92. Match the following column I with column II and choose the **correct** option.

	Column I		Column II
a.	Flagellated protozoan	(i)	They have cavity that opens to outside of the cell surface
b.	Ciliated protozoan	(ii)	It has false feet to capture its prey
c.	Amoeboid protozoan	(iii)	The parasitic form cause diseases, like sleeping sickness
d.	Sporozoan	(iv)	It includes organism that have infectious spore like stage

(1) a(iii); b(iv); c(i); d(ii)

(2) a(ii); b(i); c(iv); d(iii)

(3) a(iii); b(i); c(ii); d(iv)

(4) a(ii); b(iii); c(iv); d(i)

93. Cyanobacteria **do not** have

(1) Photosynthetic pigments

(2) Oxygenic photosynthesis

(3) Flagella throughout life cycle

(4) Cell wall surrounding the cells

94. Sac fungi are similar to `fungi imperfecti' in

(1) Absence of sexual reproduction

(2) Presence of conidia as non motile asexual spores

(3) Type of fruiting body

(4) Prolonged dikaryophase

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95. Identify the plant shown in the below given diagram and choose the statement that is **not** true for it.



(1) It belongs to a division which includes first terrestrial plants to possess vascular tissues.

(2) Its sporophyte bears sporangia that are present on sporophylls called macrophylls.

(3) It is found in cool, damp, shady places and sex organs antheridia and archegonia are produced on the main plant body, *i.e.,* sporophyte.

(4) It is frequently grown as ornamentals and has large leaves.

96. Mark True **(T)** or False **(F)** for the following statements and select the **correct** option.
A : The sporophyte in *Polytrichum* is more developed than in *Riccia*.
B : In angiosperms, ovules remain enclosed within the ovary wall.
C : *Fucus* exhibits oogamous reproduction.

	A	B	C
(1)	T	T	F
(2)	F	T	F
(3)	F	F	T
(4)	T	T	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

97. Plant body is differentiated in frond, stipe and hold fast in

(1) *Porphyra*

(2) *Laminaria*

(3) *Ulothrix*

(4) *Chlamydomonas*

98. **Statement-A** : In some leguminous plants, the leaf base swells and is called pulvinus.
Statement-B : Pulvinus produces a cooling effect in the leaves by bringing fresh air to the leaf surface.

(1) Statement-A is incorrect

(2) Statement-B is incorrect

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

99. Identify the **correct** set of statements.
(a) A sterile stamen is called a staminode.
(b) Calyx is usually brightly coloured to attract insects for pollination.
(c) In bean flowers, the standard petal overlaps the two smallest anterior petals which in turn overlap the wings.
(d) Mustard has parietal placentation.
(e) Potato family is widely distributed in tropics, subtropics but not in temperate zones.
Choose the **correct** answer from the options given below.

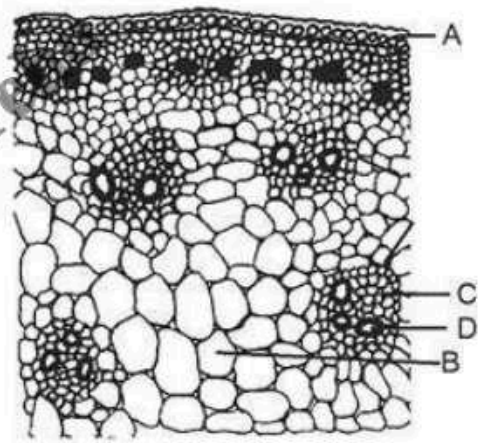
(1) (a), (b), (c) and (e) only

(2) (a) and (d) only

(3) (b) and (e) only

(4) (b), (c), (d) and (e) only

100.



Select the **correct** option w.r.t the above labelled diagram.

(1) 'A' is constituted by the collenchymatous cells

(2) 'B' is constituted by thin-wall cells made up of lignin

(3) 'C' lacks phloem parenchyma

(4) 'D' is xylem which is exarch in condition

101. Guard cells
a. Enclose stomatal pore
b. Are dumb-bell shaped in dicots
c. Do not have chloroplast
d. Regulate opening and closing of stomata.
Identify the **incorrect** statements.

(1) a & b

(2) a & d

(3) b & c

(4) c & d

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102. Vascular bundles in leaves are

(1) Conjoint and open

(2) Conjoint and closed

(3) Radial and open

(4) Radial and closed

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103. The structure of centriole differs from cilia as the former
a. Has cartwheel like appearance
b. Has 9 + 2 arrangement of axonemal microtubules
c. Is involved in the formation of spindle microtubules
d. has central proteinaceous region called hub
e. Is surface structure involved in cell motility
The correct ones are

(1) a, c and e only

(2) b, d and e only

(3) a, c and d only

(4) a, d and e only

104. The bacterial envelope consists of three layered structure. Which layers(s) is/are structurally similar to that of the eukaryotes?

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(1) Innermost layer and middle layer

(2) Innermost layer

(3) Outermost layer

(4) Middle layer and outermost layer

105. DNA content in a meiocyte at G_2 phase is 40 pg. What would be the DNA content after meiosis-II in the daughter cells?

(1) 20 pg

(2) 5 pg

(3) 10 pg

(4) 15 pg

106. All of the following statements are true for neuroglia, **except**

(1) They are the specialised cells found in brain and spinal cord

(2) More than 50% of the volume of neural tissue is composed of neuroglial cells

(3) These cells have different shapes and sizes and are thought to separate and insulate adjacent neurons.

(4) They do not support and protect the neurons

107. The birth canal is formed by

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(1) Womb along with vagina

(2) Uterus along with cervix

(3) Labia minora along with vagina

(4) Cervical canal along with vagina

108. Which is **not** correct about enzyme activity?

(1) There is an obligatory formation of an 'ES' complex in every enzyme catalyzed reaction.

(2) If 'product' is at a lower level than the 'substrate', then one need not to supply energy in order to form the product.

(3) If product is at high energy level than the substrate, then the reaction is exothermic in nature.

(4) Enzymes bring down activation energy, making the transition of 'substrate' to 'product' more easy.

109. Choose the neutral amino acid among the following which is also considered as the simplest amino acid.

(1) Valine

(2) Glycine

(3) Proline

(4) Tyrosine

110. A six carbon sugar with aldehyde group is

(1) Ribose

(2) Deoxyribose

(3) Glucose

(4) Fructose

111. The ash obtained from a tissue may contain how many of the following elements?

Sodium, Potassium, Calcium, Magnesium, Carbon, Oxygen, Hydrogen

(1) 7

(2) 6

(3) 3

(4) 4

112. Select the incorrect statement w.r.t. lipids.

(1) Some lipids have phosphorous and phosphorylated organic compound in them.

(2) Phospholipids are present in the cell membrane.

(3) Neural tissue have lipids with more complex structure.

(4) Glycerol also known as trihydroxy propane is a complex lipid.

113. Succinic dehydrogenase is inhibited by malonate. What would be the effect of this inhibition on K_m and V_{max} on increasing substrate concentration?

(1) V_{max} constant, K_m increases

(2) V_{max} constant, K_m decreases

(3) V_{max} increases, K_m increases

(4) V_{max} decreases, K_m constant

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114. If a healthy person is running fast, then
a. Every 100 mL of oxygenated blood will deliver more than 5 mL of oxygen to tissues.
b. Duration of inspiration will reduce.
c. Cardiac output will increase.
Choose the **correct** option.

(1) Only a and b are correct

(2) Only c is correct

(3) Only b and c are incorrect

(4) a, b and c are correct

115. In case of a standard electrocardiogram, the heart rate of a healthy individual can be determined by counting the number of _____, representing ventricular depolarisation, that occur in a given time period. Select the option that correctly fills the blank.

(1) P-waves

(2) T-waves

(3) QRS complexes

(4) PQ intervals

116. Diplotene stage is characterised by
(a) Dissolution of synaptonemal complex.
(b) Terminalisation of chiasmata.
(c) Complete disappearance of nucleolus.
(d) Appearance of chiasmata.
(e) Disintegration of nuclear envelope.
Choose the **correct** option.

(1) (a), (b) & (c)

(2) (c), (d) & (e)

(3) (a) & (d)

(4) All except (a)

117. Condensation of chromatin material begins in

(1) Anaphase I

(2) Prophase

(3) Metaphase I

(4) Telophase

118. State the following statements as True **(T)** or False **(F)** regarding the cell membrane and select the **correct** option.
(a) In human beings, the membrane of the erythrocyte has approximately 40 percent lipids and 52 percent proteins.
(b) Integral and peripheral proteins lie on the surface of the membrane.
(c) Quasi-fluid nature of lipids enables lateral movement of proteins within the overall bilayer.
(d) The fluid nature of the membrane helps in functions like cell growth, cell division and formation of intercellular junctions.

(1) (a)-T, (b)-T, (c)-F, (d)-T

(2) (a)-T, (b)-F, (c)-T, (d)-T

(3) (a)-T, (b)-F, (c)-F, (d)-F

(4) (a)-F, (b)-T, (c)-T, (d)-T

119. Read the following statements of Assertion (A) and Reason (R).
Assertion(A): Hydrogen ions that are produced by the splitting of water accumulate within the lumen of the thylakoids.
Reason (R): Splitting of the water molecules takes place on the inner side of the thylakoid membrane.
In the light of the above statements, choose the **correct** option.

(1) Both (A) and (R) are false statements

(2) (A) is true but (R) is false statement

(3) Both (A) and (R) are true and (R) is the correct explanation of (A)

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

120. Bundle sheath cells are similar to mesophyll cells as both

(1) Have grana

(2) Are involved in CO_2 fixation

(3) Have thick-walled cells

(4) Are impervious to gaseous exchange

121. With respect to apical dominance, which of the following pairs of PGRs act antagonistically to each other?

(1) Cytokinin and abscisic acid

(2) Auxin and cytokinin

(3) Auxin and abscisic acid

(4) Cytokinin and ethylene

122. RQ value for tripalmitin is

(1) More than that of glucose

(2) Less than 1

(3) More than that of proteins

(4) Equal to 1

123. The coenzyme A is used during the conversion of

(1) Acetyl Coenzyme A into pyruvate

(2) Succinyl coenzyme A into succinic acid

(3) Malic acid to oxaloacetic acid

(4) α-ketoglutarate into succinyl coenzyme A

124. I. In pineapples, flowering and fruit set up are synchronised by ___A___.
II. Ethylene increases number of ___B___ flowers and fruits in cucumber.
Complete the above statements by choosing correct option for A and B.

(1) **A–** Gibberellins, **B–** Female

(2) **A–** Ethylene, **B–** Female

(3) **A–** Auxin, **B–** Male

(4) **A–** Ethylene, **B–** Male

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125. *Citrus*, seeds

(1) May have genetically identical embryos

(2) Can never be apomictic

(3) Have coleorhiza as the covering of radicle

(4) Contain haploid embryos only

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126. Nucellus is **not** completely consumed and remains inside the seed of

(1) Maize

(2) Barley

(3) Beet

(4) Pea

127. "Seeds offer several advantages to angiosperms". How many of the given statements is/are **correct** in reference to this statement?

a. Seeds help the species to colonise in other areas.

b. They have sufficient food reserve, young seedlings are nourished until they are capable of photosynthesis on their own.

c. They do not have better adaptive strategies for dispersal to new habitats.

d. Hard seed coat provides protection to the young embryo.

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(1) 4

(2) 3

(3) 2

(4) 1

128. Choose the **incorrect** match.

	Blood group of offspring	Genotype of offspring	Blood groups of both parents
(1)	AB	I^AI^B	A and B
(2)	A	I^AI^O	A and AB
(3)	B	I^BI^B	A and AB
(4)	O	I^OI^O	A and A

(1) (1)

(2) (2)

(3) (3)

(4) (4)

129. A couple have two daughters and two sons. Among two daughters one is normal and other is carrier for haemophilia. Among two sons one is normal and other is haemophilic. What would be the genotypes of the parents?

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(1) XX and XY

(2) XX and X^hY

(3) XX^h and XY

(4) XX^h and X^hY

130. How many types of gametes can be produced by a diploid organism, if it is heterozygous for two loci?

(1) 2

(2) 1

(3) 8

(4) 4

131. Match **Column I** with **Column II** w.r.t. respiratory volumes and their values.

	Column I		Column II
a.	Tidal Volume	(i)	1100-1200 mL
b.	Inspiratory Reserve Volume	(ii)	1000-1100 mL
c.	Expiratory Reserve Volume	(iii)	2500-3000 mL
d.	Residual Volume	(iv)	500 mL

Choose the **correct** option.

(1) a(iv), b(ii), c(iii), d(i)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iv), b(iii), c(ii), d(i)

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132. Which of the following options is true about smooth muscles of urinary bladder and urethral sphincter causing micturition in a normal adult human?

	Smooth muscles of urinary bladder	Urethral sphincter (simultaneously)
(1)	Relax	Contract
(2)	Contract	Relax
(3)	Relax	Relax
(4)	Contract	Contract

(1) (1)

(2) (2)

(3) (3)

(4) (4)

133. During dehydration, if ADH is not secreted from hypothalamus, then it may lead to

(1) More reabsorption of Na^+ from distal tubules

(2) More loss of K^+ in urine

(3) Decrease in volume of blood

(4) More reabsorption of water from DCT

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134. Below given are the features of 'X' involved in the regulation of respiration.
(a) Neural signals from this centre can reduce the duration of inspiration.
(b) It can moderate the functions of respiratory rhythm centre.
(c) It is present in the pons region of the brain.
Select the option that correctly identifies 'X'.

(1) Pneumotaxic centre

(2) Respiratory rhythm centre

(3) Chemosensitive area

(4) Appetite centre

135. Which of the following pairs is **incorrectly** matched in humans?

(1) Gliding joint – Between zygapophyses of the successive vertebrae

(2) Fibrous joint – In the skull

(3) Hinge joint – Elbow joint

(4) Cartilaginous joint – Between the radius and ulna

136. Identify the structures labelled as X, Y and Z in the given figure.



Subunit of which of the following binds to the calcium to expose myosin-binding site on actin filament?

(1) X only

(2) Y and Z

(3) X and Z

(4) Z only

137. Hypothalamus releases a hormone 'X' which stimulates the pituitary gland to synthesise and release 'Y'. Now, 'Y' majorly acts on the Leydig cells and stimulates the synthesis of hormones from gonads.
Identify the **correct** option w.r.t. the primary and direct role of 'Y' in the human body.

(1) Induces ovulation and maintains corpus luteum

(2) Regulates the development of secondary sexual characteristics

(3) Acts on mammary glands and stimulates the formation of alveoli.

(4) Maintains the functions of male sex accessory ducts

138. Consider the following symptoms which were observed in a child
(a) Growth retardation
(b) Low IQ
(c) Abnormal skin
(d) Deaf mutism
Above mentioned symptoms are caused by deficiency of a hormone, whose hypersecretion causes

(1) Simple goitre

(2) Exophthalmic goitre

(3) Myxedema

(4) Cretinism

139. Complete the analogy by selecting **correct** option.
Internal fertilization : *Periplaneta* :: External fertilization : _____

(1) *Fasciola*

(2) *Wuchereria*

(3) *Euspongia*

(4) *Ctenoplana*

140. Select the option with **incorrect** statement w.r.t. Cyclostomata.
a. They have 6-18 pairs of gill slits for respiration.
b. Their larvae, after metamorphosis, return to fresh water.
c. Cranium and vertebral column are ossified.
d. Cyclostomes have terminal mouth with jaws.

(1) a and b only

(2) b, c and d only

(3) a, c and d only

(4) a, b, c and d

141. Female accessory ducts participating in reproduction do **not** include

(1) Oviducts

(2) Uterus

(3) Vagina

(4) Epididymis

142. Cerebral aqueduct is a canal which connects the cavity of diencephalon to cavity of medulla oblongata and passes through

(1) Cerebrum

(2) Midbrain

(3) Cerebellum

(4) Hypothalamus

143. Spermathecae in female cockroach are present in

(1) 4th abdominal segment

(2) 6th abdominal segment

(3) 5th abdominal segment

(4) 7th abdominal segment

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144. Choose the **incorrect** statement w.r.t. the reproductive structures/reproduction in frogs.

(1) A mature female can lay about 2500 to 3000 unfertilized eggs at a time.

(2) Post-fertilisation development involves a larval stage.

(3) 10-12 vasa efferentia arise from testes of males.

(4) The ovaries in female frogs are functionally connected with kidneys.

145. Read the following statements and choose the **correct** option.

Statement A : In dorsoventrally flattened worms, removal of nitrogenous wastes occurs through flame cells.

Statement B : Some members of the phylum Platyhelminthes such as *Planaria* possesses high regeneration capacity and live within alimentary canal of man.

(1) Both statements A and B are incorrect

(2) Both statements A and B are correct

(3) Statement A is incorrect but B is correct

(4) Statement A is correct but B is incorrect

146. **Assertion (A):** Spermatids transform into specialized spermatozoa during spermiogenesis without undergoing further meiotic or mitotic divisions.

Reason (R): During spermiation, spermatids undergo differentiation including condensation of nuclear material and mitochondrial rearrangement to form functional spermatozoa.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) (A) is true but (R) is false

(3) Both (A) and (R) are false

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

147. Choose the **correct** option w.r.t. a bioreactor.

(1) Large vessel of about 100-1000 litres for biomass culture

(2) A reaction centre which always consumes heat and has no temperature control system

(3) An atomic reactor to synthesize biomass on small scale

(4) It is used only for anaerobic fermentation

148. Plasmid is

(1) A part of the chromosomal DNA of bacteria

(2) Always linear and double-stranded

(3) Always devoid of antibiotic-resistance genes

(4) Capable of autonomous replication

149. Choose the **incorrect** match w.r.t. human females.

(1)	Secondary oocyte	–	Formed along with 1st polar body after first meiotic division
(2)	Theca layer	–	Organised into two layers in secondary oocyte
(3)	Primary follicle	–	Primary oocyte surrounded by a layer of granulosa cells
(4)	Tertiary follicle	–	Characterised by the presence of antrum

(1) (1)

(2) (2)

(3) (3)

(4) (4)

150. In a human female, menstruation can be deferred by the administration of

(1) FSH

(2) LH

(3) Relaxin

(4) Progesterone

151. If a foreign DNA is ligated at *Pst* I site in the vector pBR 322 then all the transformants will

(1) Grow in ampicillin containing medium

(2) Grow in tetracycline containing medium

(3) Grow in both ampicillin and tetracycline containing medium

(4) Do not grow in both ampicillin and tetracycline containing medium

152. Sieving effect for separation of DNA fragments in gel electrophoresis is provided by

(1) Agarose gel

(2) Negative charge on DNA molecules

(3) Charge on electrodes

(4) Positive charge of DNA molecules

153. Which of the following are direct methods of gene transfer?

(a) Electroporation
(b) Chemical mediated gene transfer
(c) Retrovirus mediated gene transfer
(d) Biolistics

Choose the **correct** answer.

(1) (a), (b) and (c)

(2) (a), (b) and (d)

(3) (a), (c) and (d)

(4) (c) and (d)

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154. The vector used for transfer of nematode specific genes into the host plant for mediating RNA interference is

(1) *Meloidogyne incognita*

(2) *Nicotiana tabaccum*

(3) *Penicillium expansum*

(4) *Agrobacterium tumefaciens*

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155. Biofortified crop like Golden rice developed transgenically is nutritionally rich in _____.
Select the option which fills the blank **correctly**.

(1) Vitamin B

(2) Vitamin K

(3) Vitamin D

(4) Vitamin A

156. Read the statement (A-D) and select the **correct** ones w.r.t. Morgan experiments on linkage
(A) When genes were grouped on the same chromosome, some genes were loosely linked and showed very low recombination.
(B) The genes for white eye colour and yellow body colour were tightly linked and showed 1.3 percent recombination.
(C) The genes for white eye colour and miniature wings were loosely linked and showed 37.2% parental types.
(D) Morgan found that proportion of parental gene combination was much higher than non-parental gene combination.

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(1) (A) and (C)

(2) (B), (C) and (D)

(3) (B) only

(4) (B) and (D)

157. In each human pregnancy there is always __A__ probability of a female child.
Fill the blank with suitable option.

(1) 75%

(2) 50%

(3) 25%

(4) 100%

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158. Observe the given tRNA



Select the option that cannot be true regarding the above tRNA?

(1) This tRNA will bind at AUG codon of mRNA

(2) This tRNA binds with amino acid methionine at its 3' end

(3) The loop with UAC is anticodon loop of this tRNA

(4) The amino acid X would be the same when anticodon UAC is changed to UAA

159. What will happen if the DNA isolated from virulent bacteria is kept in a test tube where non-virulent bacteria are growing?

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(1) Isolated DNA will be replaced by RNA

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(2) Isolated DNA will be rapidly degraded into nucleotides

(3) Isolated DNA will form separate chromosome in non-virulent form

(4) There may be incorporation of DNA in genome of non-virulent bacteria

160. Repressor of *lac* operon

(1) Is synthesised by *lac* z gene

(2) Binds to beta-galactosidase

(3) Exhibits negative regulation of *lac*-operon

(4) Allows RNA polymerase to access to the promoter

161. Severo Ochoa enzyme is

(1) Polypeptide phosphorylase

(2) RNA dependent DNA polymerase

(3) Polynucleotide phosphorylase

(4) DNA dependent DNA polymerase

162. All of the following are found in the structure of deoxyadenosine, **except**

(1) N-glycosidic linkage

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(2) Phosphate group

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(3) Pentose sugar

(4) Purine

163. Alexander Fleming discovered the first antibiotic while working on

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(1) *Staphylococci*

(2) *Lactobacillus*

(3) *Trichoderma*

(4) *Aspergillus*

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164. *Monascus purpureus* is a yeast used commercially in the production of

(1) Blood cholesterol lowering statins

(2) Ethanol

(3) Streptokinase for removing clots from the blood vessels

(4) Citric acid

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165. Select **odd** one out w.r.t. exponential growth model.

(1) It exhibits sigmoid curve

(2) This type of growth model is shown by r-selected species.

(3) It is observed when food and space are unlimited

(4) In this type of growth, each species has the ability to realise fully its innate potential to grow in number

166. Gause's principle of competitive exclusion states that

(1) Competition for the same resources excludes species having different food preferences.

(2) Larger organisms exclude the smaller ones through competition.

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(3) Two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually.

(4) More abundant species exclude the less abundant ones through competition.

167. Read the following statement (A-D).

(A) The rate of biomass production is expressed in terms of (kcal m^{-2})yr^{-1}.

(B) Primary productivity depends on availability of nutrients and photosynthetic capacity of plants.

(C) Primary productivity does not depend on the plants species inhabiting a particular area.

(D) Net primary productivity is the available biomass for the consumption to carnivores only.

In the light of above statement, select the **incorrect** statement(s) from the option given below.

(1) Only (A), (B) and (C)

(2) Only (B), (C) and (D)

(3) Only (C) and (D)

(4) Only (A) and (C)

168. If 500 kJ of energy is trapped at first trophic level, then how much energy will be available to snake in the following food chain?

Plants → Grasshopper → Frog → Snake

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(1) 50 kJ

(2) 5 kJ

(3) 0.5 kJ

(4) 0.05 kJ

169. Which of the given species is not **correctly** matched with its place of extinction?

(1) Dodo – Mauritius

(2) Steller's sea cow – Australia

(3) Quagga – Africa

(4) Thylacine – Australia

170. Find the **odd** one out w.r.t. alien species posing threat in India.

(1) Carrot grass

(2) Water hyacinth

(3) *Clarias gariepinus*

(4) Cichlid fish

171. The use of microorganism metabolism to remove pollutants is called

(1) Molecular diagnosis

(2) Bio weapon

(3) Biofertilizer

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(4) Bioremediation

172. The construction of the first recombinant DNA by Stanley Cohen and Herbert Boyer was done in the year

(1) 1980

(2) 1971

(3) 1982

(4) 1972

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173. From original seed-eating finches, many other forms with altered beaks arose, enabling them to become insectivorous and vegetarian finches. This process of evolution of different species starting from a point is

(1) Parallel evolution

(2) Adaptive radiation

(3) Convergent evolution

(4) Cyclic evolution

174. The similarity in Cu7 and Multiload 375 is that both

(1) Increase fertilising capacity of sperms

(2) Inhibit menstruation

(3) Make the cervix hostile to sperms

(4) Suppress sperm motility

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175. Select the features which are **true** for *Australopithecines*.
(a) Walked bipedally
(b) Earliest fossils obtained from Australia
(c) Man-like primates walked in East-African grasslands
(d) Existed probably 2 mya
Choose the **correct** option.

(1) (a), (c) and (d)

(2) (a), (b), (c) and (d)

(3) (b), (c) and (d)

(4) (a), (b) and (c)

176. Choose the incorrect option w.r.t assisted reproductive technologies.

(1) These procedures are very cheap and affordable, and can be performed by quacks.

(2) These procedures are invasive in nature.

(3) Emotional, religious and social factors can be the deterrents in the adoption of these procedures.

(4) The success rate of these procedures can vary depending on various factors like age, the cause of infertility, *etc*

177. During which period, dinosaurs became extinct?

(1) Permian

(2) Triassic

(3) Jurassic

(4) Cretaceous

178. **Assertion (A):** Morphine which is extracted from latex of poppy plant is very useful in patients who have undergone surgery.
Reason (R): It is very effective sedative and pain killer.
In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are false

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) (A) is true, (R) is false

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

179. Infective stage of *Plasmodium* to human is

(1) Sporozoite

(2) Merozoite

(3) Gametocyte

(4) Cryptozoite

180. Choose the **incorrect** statement w.r.t. spleen.

(1) It is responsible for differentiation of immature lymphocytes into antigen-sensitive lymphocytes.

(2) It is a large bean-shaped organ.

(3) It contains lymphocytes and phagocytes.

(4) It acts as a reservoir of erythrocytes.

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